

Estimating (k_0, k_1, k_2) from Individual Allele Frequencies

The solution space, measured

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1 The problem

PCAone's `--evaladmix` returns the kinship coefficient ϕ . That is not enough to identify a relationship, because **parent–offspring and full sibs both have $\phi = 1/4$** . Only the IBD sharing probabilities separate them: a parent and child share exactly one allele IBD everywhere $(k_0, k_1, k_2) = (0, 1, 0)$, while full sibs average $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$. Any pipeline that has to decide *which* relatives it has found, rather than merely that it has found some, needs k .

The input available is what PCAone already computes: the genotypes $g_{is} \in \{0, 1, 2\}$ and the individual allele frequencies

$$\hat{\pi}_{is} = \frac{1}{2} [V(V^\top V)^{-1}V^\top G]_{is},$$

the frequency expected at site s for individual i given their position on the top $K - 1$ PCs. The question is what can be built from those two objects.

This note enumerates the options, measures them on the RelateAdmix benchmark data (126 individuals, 102,185 sites, $K = 2$, a known pedigree with ten pairs per relationship class), and identifies which survive. Numbers throughout are measured, not estimated.

2 Two framing decisions

Before comparing estimators, two choices narrow the space considerably.

2.1 Estimate k only for pairs above a kinship threshold

k is meaningless for unrelated pairs — the answer is $(1, 0, 0)$ and everything else is noise. Screening on the kinship that has already been computed, at $\phi > 0.05$:

threshold	pairs kept	related kept	unrelated leaked
$\phi > 0.02$	53	53	0
$\phi > 0.05$	50 of 7875 (0.63%)	50	0
$\phi > 0.08$	40	40	0

Every first cousin and closer survives; no unrelated pair leaks through. Second cousins (true $\phi = 0.0156$) fall below, which is the right call.

The consequence is architectural, not cosmetic. **The number of related pairs is $O(N)$, not $O(N^2)$** , so the per-pair budget becomes essentially unlimited — a full likelihood is affordable — and the $O(N^2)$ memory that would otherwise cap the analysis never arises.

2.2 Relatives in admixed data do not share ancestry

This is the fact that eliminates the most obvious solution. In the benchmark data, relatives are deliberately paired across the ancestry gradient, because that is the case these methods exist to handle:

	duplicate	full sib	parent–offs.	1st cousin	half sib	2nd cousin
mean $ q_i - q_j $	0.000	0.003	0.210	0.325	0.429	0.430

A half-sib pair can differ by 0.43 in admixture proportion. So $\hat{\pi}_{is}$ and $\hat{\pi}_{js}$ genuinely differ, and any method that collapses them to a single number discards precisely what the individual allele frequencies encode. Note that the threshold of §2.1 does *not* help here: related pairs differ in ancestry *more* than unrelated ones, not less.

3 The solution space

3.1 (A) Method of moments — the PC-Relate estimators

Three statistics computed independently from the same data, not derived from one another. k_2 comes from a dominance coding, whose covariance isolates IBD-2 because it vanishes under IBD-1 sharing; k_0 from opposing-homozygote counts, which are impossible if the pair shares any allele IBD; k_1 by subtraction.

$$\hat{k}_2 = \frac{\sum_s \tilde{D}_{is} \tilde{D}_{js}}{\sum_s \hat{\mu}_{is} \hat{q}_{is} \hat{\mu}_{js} \hat{q}_{js}}, \quad \hat{k}_0 = \frac{\sum_s [\mathbf{1}(g_{is}=2, g_{js}=0) + \mathbf{1}(g_{is}=0, g_{js}=2)]}{\sum_s [\hat{\mu}_{is}^2 \hat{q}_{js}^2 + \hat{q}_{is}^2 \hat{\mu}_{js}^2]}, \quad \hat{k}_1 = 1 - \hat{k}_0 - \hat{k}_2.$$

Streams. Every accumulator is a cross-product of an $N \times m$ matrix, so each is a sum over site blocks. Verified against an explicit per-pair loop to 7.8×10^{-16} (k_2) and 1.1×10^{-15} (k_0). Memory is $O(N^2)$, independent of the number of sites.

Weaknesses. The estimators are unconstrained, and **68.4% of pairs land outside the simplex** — mostly small negative $\hat{k}_2$, since for unrelated pairs the truth is zero and an unconstrained estimator scatters both ways. Accuracy also depends on three post-hoc corrections (inbreeding subtraction, a small-sample regression on the PC outer products, and a swap to $\hat{k}_0 = 1 - 4\hat{\phi} + \hat{k}_2$ for distant pairs) carrying two hard-coded thresholds and fitted on the pairs the method itself judges unrelated.

3.2 (B) Maximum likelihood with a pooled frequency — fails

The obvious likelihood: per site, mix the three IBD-state genotype distributions using a single pooled frequency π_{ij} , and fit k by EM. It does not work:

	k_0	k_1	k_2	expected
parent-offspring	0.831	0.169	0.000	0, 1, 0
full sib	0.754	0.033	0.213	$\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

A parent-offspring pair is reported as 83% unrelated. The cause is §2.2: pooling $\hat{\pi}_i$ and $\hat{\pi}_j$ for a pair whose ancestries differ by 0.21 destroys the model. This option is listed only because it is the natural first attempt and its failure is not visible until it is tried.

3.3 (C) Maximum likelihood at the allele level — works

The fix is to stop modelling genotypes and model the alleles. Each individual's *own* frequency governs its *own* non-shared allele; only the IBD-shared allele S needs a pooled frequency:

$$\begin{aligned} \text{IBD} = 0 : & \quad g_i \sim \text{HWE}(\hat{\pi}_i), \quad g_j \sim \text{HWE}(\hat{\pi}_j) \text{ independent} \\ \text{IBD} = 1 : & \quad S \sim \hat{\pi}_{ij}, \quad A_i \sim \hat{\pi}_i, \quad A_j \sim \hat{\pi}_j \\ \text{IBD} = 2 : & \quad g_i = g_j \sim \text{HWE}(\hat{\pi}_{ij}) \end{aligned}$$

with $\hat{\pi}_{ij} = \frac{1}{2}(\hat{\pi}_i + \hat{\pi}_j)$, fitted by EM over the three states. Same inputs as (B), same optimiser, but the asymmetry is retained — and the result is exact where (B) was off by 0.83.

Does not stream. The objective is $\sum_s \log P(\cdot)$, and a sum of logs does not factor into cross-products. It needs per-site access for each pair. Under §2.1 this is irrelevant: 0.63% of pairs.

3.4 (D) Constrained ML on an admixture decomposition — RelateAdmix's model

The statistically principled version. An IBD-shared allele is drawn once from *one* ancestral population, so the correct likelihood conditions on q and f separately rather than on their product $\pi = \sum_k q_k f_k$. This is what RelateAdmix and NGSremix do, and it is why they are the reference methods for admixed relatedness.

Cost. PCAone does not produce q and f . It would need an NMF of $\hat{\Pi} \approx QF^\top$ (PCAngsd does this) or an external ADMIXTURE run — reintroducing a choice of K , a discreteness assumption about ancestral populations, and typically the dominant cost of the whole analysis. Option (C) approximates this model while avoiding all three.

3.5 (E) Segment-based methods

Out of scope by assumption, and unavailable here: IBD-segment detection needs genetic-map positions and usually phasing, neither of which is part of a PCA workflow.

4 Measured comparison

All on the same data, against theoretical kinship.

class (statistic)	truth	(C) ML from π	(A) moments, raw	(A) moments, corrected
duplicate (k_2)	1	1.000	0.966	0.979
parent-offspring (k_1)	1	1.000	1.005	1.000
full sib (k_0)	0.25	0.264	0.255	0.255
half sib (k_0)	0.5	0.565	0.530	0.495
first cousin (k_0)	0.75	0.794	0.764	0.738
unrelated (k_0)	1	0.999	0.987	—
inside the simplex		always	31.6% of pairs	mostly
post-hoc corrections needed		none	three	three
streams		no	yes	yes

Neither dominates. (C) is exact at the simplex boundaries — duplicates, parent-offspring, unrelated. The corrected moments are better in the interior, at half sibs and first cousins, where (C) attenuates ($k_0 = 0.565$ for a truth of 0.5).

5 Recommendation

Option (C), the allele-level ML, restricted to pairs with $\phi > 0.05$. The reasoning is not that it is uniformly the most accurate, because it is not:

- It is **exact on the cases that get acted on**. Duplicates and parent-offspring are what get removed from a study or flagged as sample swaps, and it separates PO from full sibs perfectly ($k_1 = 1.000$ versus 0.496).
- It is **always inside the simplex** by construction, where the raw moments put 68% of pairs outside it.
- It needs **no post-hoc corrections** — no hard-coded thresholds, no regression fitted on the pairs the method has itself judged unrelated.
- Under the threshold its one disadvantage evaporates. The moments' advantage is that they stream; streaming buys nothing when only 0.63% of pairs are being estimated.

5.1 The attenuation, and how to remove it

Option (C) attenuates at half sibs and first cousins — $k_0 = 0.565$ for a truth of 0.5. The cause is not the estimator but its input: $\hat{\pi}$ has partly absorbed the relatedness being measured, because the pair contributed to the regression that produced it. RelateAdmix shows the same signature for the same reason.

That cause can be attacked directly. Since $\hat{\beta}_s = (V^\top V)^{-1} V^\top g_s$, dropping the pair's two rows from *both* sides is a rank-2 downdate,

$$\hat{\beta}_s^{(-ij)} = (V^\top V - v_i v_i^\top - v_j v_j^\top)^{-1} (V^\top g_s - v_i g_{is} - v_j g_{js}),$$

so $\hat{\pi}_i$ and $\hat{\pi}_j$ can be re-formed without the pair at the cost of one $(k+1) \times (k+1)$ solve per pair and one extra axpy per site. It is the same idea as evalAdmix's leave-one-out refit of F , transposed to the PC regression.

It works:

	no leave-one-out			leave-one-out			expected
	k_0	k_1	k_2	k_0	k_1	k_2	
duplicate	0.000	0.000	1.000	0.000	0.000	1.000	0, 0, 1
parent-offspring	0.000	1.000	0.000	0.000	0.998	0.002	0, 1, 0
full sib	0.264	0.496	0.240	0.253	0.504	0.243	.25, .5, .25
half sib	0.565	0.434	0.001	0.530	0.466	0.004	.5, .5, 0
first cousin	0.794	0.204	0.002	0.742	0.255	0.003	.75, .25, 0
RMSE over 50 pairs $\times$ 3 parameters							
			0.03049	0.01737			43% lower

First cousins go from 0.794 to 0.742 against a truth of 0.75, and full sibs from 0.264 to 0.253 against 0.25. Half sibs improve from 0.565 to 0.530 but remain the worst case — so the correction removes most of the

attenuation, not all of it. Duplicates and parent–offspring are unaffected, as they should be: they were already exact.

And it is free. Timed over three runs of the implementation described in §5.2:

	wall (16 threads)	wall (1 thread)
<code>--no-loo</code>	0.83–0.86 s	6.10 s
leave-one-out (default)	0.82–0.86 s	6.03 s

Within run-to-run noise, single- or multi-threaded. The downdate adds $O(M(k+1)^2)$ per pair against the $O(M(k+1))$ it replaces — with $k+1=2$, four multiply–adds per site instead of two — which disappears beside reading the `.bed` and running the EM. There is no reason not to make it the default, and it is.

What remains. Half sibs at 0.530 for a truth of 0.5. The pair is now out of the frequency regression, but it is still in the PCA that produced the scores V ; removing it there too would mean recomputing the decomposition per pair, which is not free in the way this is. Whether that residual 0.03 is worth chasing is a separate question, and it should be measured on a cohort with more than 126 individuals before anyone tries.

5.2 Architecture: a second program

Everything needed is already in PCAone’s published output. This is the fact that decides the question, and it is not obvious — $\hat{\pi}$ looks like an internal quantity, and `--evaladmix` never actually forms it (the Gram-matrix identity lets it avoid doing so). But

$$\hat{\pi} = \frac{1}{2} V(V^\top V)^{-1} V^\top G, \quad V = [\text{PC}_1 \dots \text{PC}_{K-1}, \mathbf{1}]$$

needs only the PC scores and the genotypes — and the genotypes are needed for the likelihood regardless. So `.eigvecs` plus the `.bed` is sufficient; the singular values and loadings are not required, and neither is any PCAone internal.

Confirmed by building exactly that: a program reading only `recon.eigvecs`, `recon.kinship` and the PLINK files reproduces the estimates *identically* — duplicates (0, 0, 1), parent–offspring (0, 1, 0), full sibs (0.264, 0.496, 0.240), matching the in-process prototype to three decimals.

Given that, a separate program is the better structure:

- The work is a **different computational shape**: a per-pair EM over a sparse list of $O(N)$ pairs, not streaming matrix algebra over sites. It shares no machinery with PCAone’s inner loop beyond reading genotypes.
- It adds **no new $N \times N$ state** to PCAone and no second pass to its control flow.
- Anyone with existing PCAone output can run it **without redoing the PCA**.
- **The statistics are not settled** — the attenuation at half sibs, the choice of $\hat{\pi}_{ij}$, whether a leave-one-out refinement helps. Iterating on those should not require rebuilding and reshipping a PCA tool. The working prototype above is about forty lines.

This reverses the recommendation in the first draft of this note, which argued for a second pass inside PCAone. That argument rested on the assumption that $\hat{\pi}$ was only available internally. It is not.

One change to PCAone is required. The second program must apply *exactly* the site filter the PCA used, or $\hat{\pi}$ will not correspond to the PCs. PCAone already solves this with `.mbim`, which lists the retained sites and their allele frequencies — but **only a run with `-D/--ld` writes one**. `--evaladmix` should call `save_snps_in_mbim()` as well. That is a one-line change and it is the only coupling needed.

Interface.

```
PCAone -b plink -k <K-1> --evaladmix -o pcs
pcaone-ibd -b plink -P pcs --min-kin 0.05 -o rel
```

reading `pcs.eigvecs`, `pcs.kinship` and `pcs.mbim`, and writing `rel.ibd`: one row per surviving pair, ID1 ID2 kin k0 k1 k2 nsnp. A sparse list rather than a matrix, which is the honest representation given that k is meaningless for the 99.4% of pairs that are unrelated.

The alternative, if a single binary is preferred. PCAone already has a two-invocation idiom via `-P/--USV` — it is how the ancestry-adjusted LD path works. A mode `PCAone -b plink -P pcs --ibd-probs` would reuse the existing genotype readers and `.mbim` handling for free, at the cost of a rebuild whenever the statistics change. Reasonable once they stabilise; the wrong order in which to do it now.

6 Open questions

- $\hat{\pi}_{ij} = \frac{1}{2}(\hat{\pi}_i + \hat{\pi}_j)$ is a choice, not a derivation. Under an admixture model the IBD allele comes from one ancestral population, which $\hat{\pi}$ alone cannot identify. The mean works well here; it is untested where ancestries are more extreme than this dataset's.
- Everything measured is $N = 126$, $K = 2$, complete data, one pedigree. The claim that related pairs are $O(N)$ is a property of real cohorts, not of this dataset; a cohort with large families would change the cost.
- The leave-one-out correction removes most of the attenuation but not all of it: half sibs remain at $k_0 = 0.530$ against a truth of 0.5. The residue comes from the pair still being inside the PCA itself, which a per-pair downdate cannot reach.
- A threshold of 0.05 excludes second cousins by construction. That is defensible — their k_2 is noise — but it should be stated in the output rather than left implicit.